

1. Administrative & Study Identification

Full Study Title: A Double-Blind, Placebo-Controlled and Active-controlled Trial to Evaluate the Effect of a Supratherapeutic Dose of MK-8527 on the QTc Interval in Healthy Adult Participants
Short Title/Acronym: A Clinical Study of MK-8527 in Healthy Adult Participants (MK-8527-014)
Protocol Number: MK-8527-014
NCT Number: NCT07063238
Sponsor Name: Merck (Merck Sharp & Dohme LLC)
Investigational Product: MK-8527 (alimatravir)
Phase of Development: Phase I (PHASE1)
Medical Monitor: Merck Sharp & Dohme LLC Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the effect of a single oral higher-than-normal (supratherapeutic) dose of MK-8527 on the QTc interval (a measure of heart rhythm) in healthy adult participants. **Secondary Objectives:** To evaluate the safety, tolerability, and pharmacokinetics of MK-8527, and the change from baseline in QTc following moxifloxacin administration (active control).

2.2 Background & Rationale To support the clinical development of MK-8527, a novel nucleoside reverse transcriptase translocation inhibitor (NRTTI) being studied as a once-monthly oral tablet for HIV-1 pre-exposure prophylaxis (PrEP). Regulatory guidelines require a thorough QT/QTc assessment for new systemically bioavailable drugs to ensure they do not carry a proarrhythmic risk by prolonging the QTc interval.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled and Active-controlled (Moxifloxacin) **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Trial Status: Active but not currently recruiting (ACTIVE_NOT_RECRUITING) **Target \$N\$:** Approximately 42 healthy adult participants **Number of Sites:** Multiple US-based clinical research centers **Estimated Study Start:** 2024 **Estimated Primary Completion:** Completed

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Healthy adult participants in good health before randomization. 2. Body mass index (BMI) between 18 and 32 kg/m², inclusive. 3. Willing to comply with study restrictions and procedures.

4.2 Exclusion Criteria 1. History of clinically significant endocrine, gastrointestinal (GI), cardiovascular, hematological, hepatic, immunological, renal, respiratory, genitourinary, or major neurological abnormalities or diseases. 2. History of cancer (malignancy). 3. Positive test(s) for Hepatitis B surface antigen (HBsAg), hepatitis C antibodies, or human immunodeficiency virus (HIV).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single oral supratherapeutic (higher-than-normal) dose of MK-8527. **Route of Administration:** Oral. **Active Control Details:** Moxifloxacin (ATC Codes: J01MA14, S01AE07; Trade Names: Moxeza, Vigamox, Avelox). **Duration of Treatment:** Single dose administration with subsequent monitoring.

5.2 Concomitant & Prohibited Therapy Permitted: Standard allowed medications per Phase 1 healthy volunteer study protocols. **Prohibited:** Concomitant medications known to prolong the QTc interval or strong CYP inducers/inhibitors during the washout and treatment periods.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Day -28 to -1	Informed Consent, Medical History, Physical Exam, ECG, Labs
2	Day 1 (per period)	Randomization, Single Dose Administration (MK-8527, Moxifloxacin, or Placebo), Serial ECGs and PK blood draws
3	Post-Dose	Continuous telemetry, safety monitoring, AE collection
4	Follow-up	Final Vital Signs, Exit Interview, Safety follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change from Baseline in QT interval corrected for heart rate (QTc) following MK-8527 administration. **Measurement Method:** Serial 12-lead Electrocardiogram (ECG) assessments.

7.2 Secondary & Exploratory Endpoints 1. Number of participants who experience one or more adverse events (AEs). 2. Number of

participants who discontinue study intervention due to an AE. 3. Change from Baseline in QTc following moxifloxacin administration (to establish assay sensitivity).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring during confinement and at follow-up visits; evaluated via patient reporting, clinician assessment, and continuous cardiac rhythm monitoring. **Serious Adverse Events (SAEs):** Reported within 24 hours to the Sponsor and Institutional Review Board (IRB). **Data Monitoring Committee (DMC):** Internal safety review team evaluating blinded/unblinded safety data throughout the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: * **Safety Set:** All participants who received at least one dose of study drug. * **PK/PD Set:** Participants with sufficient data to model the concentration-QTc relationship. **Sample Size Justification:** The sample size (N=42) is designed to provide adequate statistical power to exclude a clinically significant QTc prolongation (typically >10 ms upper bound of the 90% confidence interval). **Handling of Missing Data:** Missing QTc data handled using mixed-effects models for repeated measures (MMRM); PK parameters calculated using standard non-compartmental analysis.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) guidelines and the Declaration of Helsinki. **Monitoring Plan:** Standard onsite/remote clinical monitoring of source documents and eCRF data. **Audit Trail:** 21 CFR Part 11 compliant electronic data capture (EDC) system with full audit trails.

11. Study Identifiers & Governance

Primary ID: MK-8527-014 **Posting ID:** 892558012648913335 **Registry Number:** NCT07063238 **Sponsor/Lead Organization:** Merck (Merck Sharp & Dohme LLC) **Study Title:** A Clinical Study of MK-8527 in Healthy Adult Participants **Official Regulatory Title:** A Double-Blind, Placebo-Controlled and Active-controlled Trial to Evaluate the Effect of a Supratherapeutic Dose of MK-8527 on the QTc Interval in Healthy Adult Participants

12. Clinical Framework (MK-8527 Specific)

Primary Condition / Disease Area: Healthy Volunteers **Preferred UMLS Name:** Healthy subject **Ontology Classifications:** SNOMED ID: 69830002 (Healthy finding); HPO Code: HP:0032322 **Diagnosis Threshold:** Healthy without HIV **Medical Methodology:** Prophylactic antiretroviral administration **Optimization Target:** Confirming cardiac safety profile (QTc interval) for monthly oral dosing

13. Study Procedures & Methodology

Management Pathway: Healthy Volunteer Phase 1 Pathway **Education Protocol (HCP):** Investigator site training on protocol, exact ECG timing, and sample handling **Education Protocol (Patient):** Informed consent process, PrEP overview, and visit schedule orientation **Quality Audit Mechanism:** Centralized ECG reading and independent safety reviews

14. Observation & Follow-Up Schedule

Total Duration: Dependent on the number of crossover periods and washout (approx. 4-8 weeks) **Onsite Visit Cadence:** Confinement periods surrounding each dose administration **Remote Monitoring Frequency:** Follow-up safety calls post-discharge **Checklist Review Interval:** Daily during confinement for AEs

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				